Efficient estimation of Independent Component Analysis Systems with characteristic functions

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November 11, 2026

Abstract

Although approaches to Independent Component Analysis (ICA) based on characteristic function seem theoretically elegant, they may suffer from implementational challenges because of numerical integration steps or selection of tuning parameters. Extending previously considered objective functions and leveraging results from the continuum Generalized Method of Moments of Carrasco and Florens (2000), I derive an optimal estimator that can take a tractable form and thus bypass these concerns. The method shares advantages with characteristic function approaches – it does not require the existence of higher-order moments or parametric restrictions – while retaining computational feasibility and asymptotic efficiency. The results are adapted to handle a possible first step that delivers estimated sensors. Finally, a by-product of the approach is a specification test that is valuable in many ICA applications. The method's effectiveness is illustrated through simulations, where the estimator outperforms efficient GMM, JADE, or FastICA, and an application to the estimation of Structural Vector Autoregressions (SVAR), a workhorse of the macroeconometric time series literature.

Keywords: Independent Component Analysis, characteristic function, continuum GMM, Structural VAR.

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[†]I thank Susanne Schennach for many helpful comments. I also thank Alessio Moneta, and seminar participants at Brown for constructive discussions while presenting an early version of the paper.

1 Introduction

Independent Component Analysis (ICA; Comon (1994); Eriksson and Koivunen (2003*b*)) is a popular method which finds applications in fields as diverse as signal processing, machine learning, or Structural Vector Autoregressions (SVAR).

The standard model posits that an observed vector (of "sensors") at time t, η_t, is generated through $\eta_t = \Theta\varepsilon_t$ where ε_t has independent entries and Θ is an unknown $n \times n$ matrix. ε_t, sometimes referred as the "sources", is a vector that contains the latent factors that affect the system through the mixing matrix, Θ.

Various methods have been proposed to uncover the unmixing matrix, Θ^{-1}. Early methods typically attempted to maximize a measure of non-normality or use maximum entropy, often making use of third- and fourth-order moments. A popular and fast method based on non-Gaussianity is the fastICA algorithm (Oja and Yuan, 2006). A list of algorithms and applications can be found in Hyvärinen, Karhunen and Oja (2001); Agrawal, Gupta and Garg (2022).

A broad estimation strategy is to rely on maximum likelihood or related methods. Many papers (Bach and Jordan, 2002; Samarov, Tsybakov et al., 2004; Chen, Bickel et al., 2006; Ilmonen, Paindaveine et al., 2011; Samworth, Yuan et al., 2012; Ablin, Cardoso and Gramfort, 2018) have assumed parametric, smooth, or log-concave densities to devise an estimation strategy for the unmixing matrix. Nevertheless, misspecification bias is a concern as the family of distributions is typically unknown, and assumptions of smoothness, unimodality, or absence of atoms are not innocuous in applications. In addition, many of these approaches require a choice of tuning parameter or are not straightforward to implement. Efficient approaches to ICA remain relatively rare, although a few have been proposed (*e.g.*, Chen, Bickel et al. (2006)).

I propose a nonparametric approach to estimate the unmixing matrix based on the empirical characteristic function that does not require the existence of higher-order moments or distributional restrictions and can achieve asymptotic efficiency. I also explicitly allow η_t to be (consistently) estimated rather than directly observed to account for vanishing noise or an estimation step, as happens, for instance, in Structural Vector Autoregressions (SVAR) applications where the ICA system requires inputs from a first-step regression.

Eriksson and Koivunen (2003*a*) (see also Chen and Bickel (2005) for asymptotic

properties) previously obtained an identifying equation function in terms of the characteristic function and proposed an integration scheme to estimate ICA systems. This paper shows that refining their objective function enables the derivation of an optimal weighting scheme, leading to an efficient estimator that can bypass numerical integration steps, is compatible with noise due a first step such as autoregressions in SVAR, and delivers a test of the ICA system's validity.

2 Estimation of ICA systems

2.1 Identification of Θ

Identification of systems of linear combinations of unobserved independent variables has been extensively discussed in the literature and extended to more general systems (Reiersøl, 1950; Comon, 1994; Bonhomme and Robin, 2009; Ben-Moshe, 2016). The ICA system remains a workhorse theoretical model and is routinely used in empirical applications. A general result on the identification of such systems is provided by Eriksson and Koivunen (2003*b*).

In the square matrix case, identification is easily established by noting that two observationally equivalent systems (Θ, ε) and $(\Theta^*, \varepsilon^*)$ must satisfy $\Theta\varepsilon \overset{d}{=} \Theta^*\varepsilon^*$ and thus $\varepsilon \overset{d}{=} \Theta^{-1}\Theta^*\varepsilon^*$, where $\overset{d}{=}$ denotes equality in distribution. But if ε is to have independent entries, the Darmois–Skitovich theorem (Darmois, 1953; Skitovitch, 1953) requires trivial linear combinations in the absence of normality. As a result, $\Theta^{-1}\Theta^*$ must be a permutation matrix (possibly scaled). This well-known result is summarized in the following theorem:

Theorem 1 (Identification)**.** *Consider the system $\eta_t = \Theta\varepsilon_t$. Θ is identified up to column scale and permutations from the distribution of η_t if (i) it is invertible, and (ii) the vector ε_t contains independent random variables among which at most one is normal.*

Since identification is obtained only up to scale and column permutations, a normalization and an order are still needed. In the absence of application-specific knowledge to assign identities to shocks, the choice can be made out of convenience.

I use a unit norm normalization for each column of Θ and denote the corresponding compact parameter space by $\overline{\Theta}$. The constraint can be explicited through polar

coordinates. For instance, in the 2×2 case, $\Theta = \begin{pmatrix} \cos(\theta_1) & \cos(\theta_2) \\ \sin(\theta_1) & \sin(\theta_2) \end{pmatrix}$, where θ_1 and θ_2 lie between $-\pi/2$ and $\pi/2$. Columns can be easily ordered by setting, say, $\theta_1 < \theta_2$, with straightforward adaptations to higher-dimensional settings using lexicographic ordering. The sign convention can be imposed *via* $\Theta_{1k} > 0 \forall k$ (moving to the next row in case of an exact 0). As a result, the properties can equivalently be discussed in terms of θ, under the bijective mapping $\Theta = \Theta(\theta)$.

In what follows, it is assumed that the system is identified by imposing conditions (i) and (ii) of Theorem 1 and the normalizations described above.

2.2 The class of estimators based on characteristic functions

I will make use of the following notation. I define $\vec{s}$ to be a $1 \times n$ row vector. φ_X denotes the characteristic function of the random vector X, *i.e.* $\varphi_X(\vec{s}) \stackrel{\text{def}}{=} \mathbb{E}[e^{i\vec{s}x}]$. φ (without a subscript) refers to the characteristic function of η. Real and imaginary parts are denoted by $\Re$ and $\Im$, respectively. The jth column of Θ is an $n \times 1$ vector denoted by $\Theta_{\cdot j}$.

2.2.1 The estimator

By independence of sources, the observed variables' distribution is related to the distribution of their unobserved counterparts through

$$\varphi(\vec{s}) = \prod_{j=1}^{n} \varphi_{\varepsilon_j}(\vec{s}\Theta_{\cdot j}) \tag{1}$$

while each source's characteristic function can be recovered from that of the sensors via

$$\varphi_{\varepsilon_j}(s) = \varphi(s\Theta_{j\cdot}^{-1}) \tag{2}$$

where $\Theta_{j\cdot}^{-1}$ is the j-th row in Θ^{-1}.

A functional equation for the characteristic function of η in terms of the unknown Θ can be obtained using the last two expressions. First, define $P_j \stackrel{\text{def}}{=} \Theta_{\cdot j}\Theta_{j\cdot}^{-1}$ whose immediate properties are $P_jP_k = \mathbb{1}_{j=k}P_j \ \forall j,k$, $\sum_{j=1}^{n} P_j = I_n = \sum_{j=1}^{n} P_j'$, and $\text{rank}(P_j) = 1$.

In addition, the collection of P_j is isomorphic to Θ once a normalization on Θ is imposed. Next, substituting (2) into (1) yields an expression which directly links the characteristic function of η to Θ:

$$\varphi(\vec{s}) = \prod_{j=1}^{n} \varphi(\vec{s}P_j) \tag{3}$$

This result was also obtained by Eriksson and Koivunen (2003*a*). The identity can be expressed as the condition $q(\vec{s},\theta) \stackrel{\text{def}}{=} \varphi(\vec{s}) - \prod_{j=1}^{n} \varphi(\vec{s}P_j) = 0$ for all $\vec{s}$.

Remark 1. If one assumes that the characteristic function does not vanish[1] or restricts the analysis to a neighborhood of the origin, then the criterion can alternatively be expressed in terms of the cumulant generating function: $\sum_{j=0}^{n} a_j \ln\left(\frac{1}{T}\sum_{t=1}^{T} e^{i\vec{s}P_j\eta_t}\right) = 0$ with $a_j = (-1)^{\mathbb{1}_{j>0}}$ and $P_0 = I$. Both criteria can be used to form an optimally weighted estimator of Θ with closely related expressions; the log form, however, requires more care to handle zeros.

Let $\hat{\varphi}(\vec{s}) \stackrel{\text{def}}{=} \frac{1}{T}\sum_{t=1}^{T} e^{i\vec{s}\eta_t}$ be the empirical counterpart of φ_η and let $q_T(\vec{s},\theta) \stackrel{\text{def}}{=} \hat{\varphi}(\vec{s}) - \prod_{j=1}^{n} \hat{\varphi}(\vec{s}P_j)$. Eriksson and Koivunen (2003*a*) consider a criterion based on minimizing integrals of the form

$$\int |q_T(\vec{s},\theta)|^2 w(\vec{s})\mathrm{d}\vec{s} = \int (\Re q_T(\vec{s},\theta)^2 + \Im q_T(\vec{s},\theta)^2) w(\vec{s})\mathrm{d}\vec{s} \tag{4}$$

using some weight function w. They then propose tractable weighting schemes to facilitate integration. The approach is neat, but suffers from a couple of shortcomings. First, it does not allow for weighting interactions at different points $\vec{s}$, which precludes efficiency because neighboring points contain similar information. Second, the weights are chosen suboptimally in order to avoid numerical integration, which can lead to further efficiency loss.

In analogy with the formation of a quadratic form for estimating equations or Generalized Method of Moments (GMM; Hansen (1982)), consider

$$Q_T(\theta) = \int\int \begin{pmatrix} \Re q_T(\vec{s},\theta) & \Im q_T(\vec{s},\theta) \end{pmatrix} W(\vec{s},\vec{r}) \begin{pmatrix} \Re q_T(\vec{r},\theta) \\ \Im q_T(\vec{r},\theta) \end{pmatrix} \pi(\mathrm{d}\vec{r})\pi(\mathrm{d}\vec{s}) \tag{5}$$

[1] Assumptions excluding zeros are common, *e.g.*, literature on nonparametric deconvolution (see Schennach (2004) and references therein).

where π is a probability measure and W is a (symmetric, positive semi-definite) weighting matrix.

This criterion can be induced by a linear operator B by considering the norm $\|Bq_T\|_H$, where H is a Hilbert space of square-integrable functions with the scalar product $< f, g >= \int f(\vec{s})\overline{g(\vec{s})}\pi(\vec{s})$. This is similar to Carrasco and Florens (2000, 2002)'s objective function for a continuum of moment conditions.[2] The following section establishes the asymptotic properties of estimators obtained by minimizing (5) for any weighting matrix. The weighting matrix W_T is allowed to be an estimate of some target matrix W. The results are also made applicable to the case where η_t is not observed but can be obtained from a first step, as in SVAR applications.

2.2.2 Asymptotics

To establish consistency, it is necessary to exclude sequences toward degenerate matrices if η_t is to be estimated. I make use of the following assumption, which slightly strengthens the invertibility assumption by bounding the matrix an ϵ away from degeneracy.

Definition 1 (ϵ-invertibility)**.** The angle between any two columns of Θ is of at least $\epsilon > 0$.

In applications where the η_t's are directly observed, consistency can be established under the usual (weaker) full-rank condition on Θ.

In what follows, θ_0 refers to the true value of θ and I let $q_0(\vec{s}) \overset{\text{def}}{=} q(\vec{s}, \theta_0)$. The proposed estimator is consistent for the true value θ_0 for any weighting scheme with a proper limit. This is summarized by the following Theorem, which is proven in the Appendix.

Theorem 2 (Consistency)**.** *The estimator* $\hat{\theta} \overset{\text{def}}{=} \arg\min_\theta Q_T(\theta)$ *is consistent for* θ_0 *if (i) there is a consistent estimator,* $\hat{\eta}_t$*, of* η_t *that satisfies* $\frac{1}{T}\sum_{t=1}^{T}|\hat{\eta}_t - \eta_t| \to^p 0$ *and* $\Theta = \Theta(\theta)$ *is* ϵ*-invertible, and (ii)* W_T *converges uniformly to a positive definite matrix* W *that satisfies* $\sup_{\vec{s},\vec{r}} W_{ij}(\vec{s}, \vec{r}) < \infty$*.*

[2] A difference is that q_T is obtained from a transformation of moments. Moment conditions can be obtained by introducing characteristic functions of the shocks as nuisance parameters. Up to negligible terms, the solution uses empirical characteristic functions and thus coincides with the nonlinear system.

I now turn to the derivation of the asymptotic distribution of the estimator. The following assumption imposes some structure on the way η_t is estimated.

Assumption 2 (First stage)**.** $\hat{\eta}_t$ is obtained from a first stage in which η_t is an error independent error term independent of regressors x_t so that $\hat{\eta}_t - \eta_t = x_t'(\hat{\beta} - \beta)$ for some estimator $\hat{\beta}$ which is consistent for β. In addition, $\Theta = \Theta(\theta)$ is ϵ-invertible.

If the η_t's are observed, the assumption can again be replaced by a simple full-rank condition on Θ. Then, the asymptotic normality result can be stated as follows:

Theorem 3 (Asymptotic Normality)**.** *$\hat{\theta}$ is asymptotically normally distributed. Specifically, if (i) the first stage assumption holds, (ii) W_T converges uniformly to a positive definite matrix W that satisfies $\sup_{\vec{s},\vec{r}} W_{ij}(\vec{s},\vec{r}) < \infty$, and (iii) ε_t has second moments.*

Then, $\sqrt{T}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0; BVB')$ where

$$B \overset{\text{def}}{=} \left[\int \left(\Re \tfrac{\partial q_0}{\partial \theta}(\vec{r}) \quad \Im(\tfrac{\partial q_0}{\partial \theta}(\vec{r}) \right) W(\vec{r},\vec{s}) \begin{pmatrix} (\Re \tfrac{\partial q_0}{\partial \theta}(\vec{s}))' \\ (\Im \tfrac{\partial q_0}{\partial \theta}(\vec{s}))' \end{pmatrix} \pi(d\vec{r})\ \pi(d\vec{s}) \right]^{-1} \tag{6}$$

and

$$\begin{aligned} V \overset{\text{def}}{=} \int \int & \left(\Re \tfrac{\partial q_0}{\partial \theta}(\vec{r}) \quad \Im \tfrac{\partial q_0}{\partial \theta}(\vec{r}) \right) W(\vec{r},\vec{s}) K(\vec{r},\vec{s}) \\ & W(\vec{r},\vec{s})' \begin{pmatrix} (\Re \tfrac{\partial q_0}{\partial \theta}(\vec{s}))' \\ (\Im \tfrac{\partial q_0}{\partial \theta}(\vec{s}))' \end{pmatrix} \pi(d\vec{r})\ \pi(d\vec{s}) \end{aligned} \tag{7}$$

where $K(\vec{r},\vec{s})$ is the covariance function for the real and imaginary parts of q_0.

As usual, the iid assumption can be weakened to ergodicity and strict stationarity. This is done with a natural adaptation of the proof, noting that results about convergence of characteristic functions have generalizations to ergodic, stationary settings (Feuerverger, 1990). Two corrections may apply to the asymptotic variance. First, the use of a long-run variance might be warranted since uncorrelatedness of sensors does not translate to that of their empirical characteristic functions. Second, estimation of η_t must be accounted for since its disappearance hinges on the vanishing of the term $\frac{\partial q_0(\eta_t(\beta))}{\partial \beta}(\hat{\beta} - \beta)$, which relies on independence.

Finally, since Θ is usually the parametrization of interest, the Delta method can be applied and asymptotic normality ensues for the corresponding estimator with the same asymptotic variance scaled by $\partial_\theta \Theta(\theta)$. The asymptotic variance is analyzed in details in Appendix B.

3 Efficient estimation

3.1 Optimal objective function

Since the asymptotic variance in Theorem 3 has the common "sandwich" form, a natural choice of weighting function consists in inverting the covariance term in order to minimize the asymptotic variance. This is reminiscent of the choice of efficient matrix in GMM (Hansen, 1982) and its extension to a continuum of moments conditions developed by Carrasco and Florens (2000).

As Carrasco and Florens (2000) establish, efficient estimation in the continuous case requires inverting the covariance operator $C : h \to \int K(\vec{r}, \vec{s})h(\vec{s})\mathrm{d}\vec{s}$ which is not possible on the whole reference space. They propose a regularized sample version $C_T^{\alpha_T}$ where α_T is a smoothing parameter that disturbs the eigenvalues of C. The choice of α_T has been discussed in subsequent papers, see Carrasco and Kotchoni (2017) and Amengual, Carrasco and Sentana (2020).

The optimal estimator minimizes

$$Q_T^* \overset{\text{def}}{=} \sum_{t=1}^{T} \frac{\mu_{t;T}}{\mu_{t;T}^2 + \alpha_T} |< q_T, \chi_{t;T} >|^2 \tag{8}$$

where $\lambda_{t;T}$ and $\chi_{t;T}$ are the eigenvalues and eigenfunctions of C_T, the sample counterpart to C. These can be easily deduced from the eigen-decomposition of a matrix M, as explained in the next subsection.

Moreover, under the assumptions of Theorem 3.2 and provided $\alpha_T \to 0$ while $\alpha_T^3 T \to \infty$, the expected simplification of the asymptotic variance occurs so that the asymptotic distribution becomes

$$\sqrt{T}(\hat{\theta} - \theta_0) \xrightarrow{d} N\left(0; \left|\left|\frac{\partial q_0}{\partial \theta}\right|\right|_C^{-2}\right) \overset{d}{=} N\left(0; B\right) \tag{9}$$

where the weighting matrix used to compute B is here based on the inverted covariance operator.

Although the objective function appears to induce a computationally intensive procedure due to multiple integrations, in practice the estimator is obtained by minimizing equation (8) and the main computational burden arises from evaluating a matrix and computing its eigen-decomposition. Furthermore, using the efficient form

of the estimator carries significant benefits. It removes the need to specify an arbitrary form of the weighting matrix and furthers efficiency. In particular, Carrasco and Florens (2000) show that the use of a continuum of moment conditions closes the efficiency gap between GMM and MLE.

The efficiency follows from results in (Carrasco and Florens, 2002) by completeness of characteristic functions, once a few regularity conditions hold:

Assumption 3 (Sufficient conditions for efficiency)**.** Each shock admits a continuously differentiable density and has finite variance. Moreover, either η_t is observed or the first stage assumption holds.

Simulations show a good behavior of the method in general, although guaranties of optimality follow from the theory only in the continuous case.

3.2 Implementation

Let Ψ^τ denote the moment conditions at observation τ. The covariance operator reads

$$\begin{aligned}(Cg)(\vec{r}) &= \int \frac{1}{T}\sum_{\tau=1}^{T} \Psi^\tau(\vec{r})\overline{\Psi^\tau(\vec{s})}' g(\vec{s})\ \pi(\mathrm{d}\vec{s}) \\ &= \frac{1}{T}\sum_{\tau=1}^{T} \Psi^\tau(\vec{r}) \int \overline{\Psi^\tau(\vec{s})}' g(\vec{s})\ \pi(\mathrm{d}\vec{s})\end{aligned}$$

which implies that the eigenfunctions take the form

$$\chi(\vec{r}) = \frac{1}{T}\sum_{\tau=1}^{T} c_\tau \Psi^\tau(\vec{r}) \tag{10}$$

The eigenvalues and coefficients are given by the eigenvalue-eigenvector pairs of M/T, where the elements of M are given by

$$M_{\tilde{\tau}\tau} = \int \overline{\Psi^\tau(\vec{s})}' \Psi^{\tilde{\tau}}(\vec{s})\ \pi(\mathrm{d}\vec{s}) \tag{11}$$

The eigenvector/eigenvalue couples (c, λ) of M/T are then used to form orthonormalized[3] eigenfunctions $\chi_{t,T}, t = 1, \ldots, T$.

[3]The eigenfunctions are automatically orthogonal, but need to be normed.

Computing (8) thus requires two main objects: the matrix M and the scalar products $< q_T, \chi_{t;T} >$. Their computation can be significantly simplified because the choice of integration measure π does not affect asymptotic properties (Carrasco and Kotchoni, 2017). Hence, as long as $\pi > 0$ over the whole space, ensuring tractability for ease of implementation is the main concern. I propose to integrate with respect to a Gaussian, *i.e.*, $\pi(\mathrm{d}\vec{s}) = (2\pi)^{-n/2} e^{-\frac{1}{2}\|\vec{s}\|^2}$. Then, one can compute M whose entries read

$$\begin{aligned}
&\int \overline{\Psi^{\tau}(\vec{s})}' \Psi^{\tilde{\tau}}(\vec{s})\ \pi(\mathrm{d}\vec{s}) \\
&= \exp(-0.5\ \|\eta_{\tau} - \eta_{\tilde{\tau}}\|^2) + \sum_{j=1}^{n} \exp(-0.5\ \|P_j(\eta_{\tilde{\tau}} - \eta_{\tau})\|^2) \\
&-2\frac{1}{T^n} \sum_{\{\tilde{\tau}_k\}} \exp\left(-0.5\ \left\|\sum_{k=1}^{n} P_k \eta_{\tilde{\tau}_k} - \eta_{\tau}\right\|^2\right) - 2\frac{1}{T} \sum_{\tilde{\tau}=1}^{T} \sum_{j=1}^{n} \exp\left(-0.5\ \|P_j(\eta_{\tilde{\tau}} - \eta_{\tau})\|^2\right) \\
&+\frac{1}{T^{2n}} \sum_{\{\tau_j, \tilde{\tau}_k\}} \exp\left(-0.5\ \left\|\sum_{k=1}^{n} P_k \eta_{\tilde{\tau}_k} - \sum_{j=1}^{n} P_j \eta_{\tau_j}\right\|^2\right) + \frac{1}{T^2} \sum_{\tau=1}^{T} \sum_{\tilde{\tau}=1}^{T} \sum_{j=1}^{n} \exp\left(-0.5\, \|P_j(\eta_{\tilde{\tau}} - \eta_{\tau})\|^2\right)
\end{aligned}$$

noting that $\int e^{\vec{s}(a+bi)} \pi(\mathrm{d}\vec{s}) = e^{\|a\|^2/2} e^{a'bi} e^{-\|b\|^2/2}$.

Similarly, one can obtain the scalar products as linear combinations of $< \Psi, \Psi^{\tau} > = \frac{1}{T} \sum_{t=1}^{T} \int \overline{\Psi^{\tau}(\vec{s})}' \Psi^{t}(\vec{s}) \pi(\mathrm{d}\vec{s})$.

Therefore, computing the objective function does not require numerical integration. The averages over all pairs of time can be high-dimensional when T and n get large, so that drawing times at random and computing the resulting sample average may be desirable to lower the computational cost.

3.3 Tests

Asymptotic normality provides the basis for usual confidence intervals and tests. Moreover, an advantage of the analogy with GMM is the potential for a specification test, in the spirit of over-identifying restrictions. As detailed in Carrasco and Florens (2000), such a test can be constructed on the basis of

$$\frac{||\sqrt{T}Q_T^*||^2_{C_T^{\alpha_T}} - \sum_{t=1}^{T} \frac{\lambda_{t;T}^2}{\lambda_{t;T}^2+\alpha_T}}{\sqrt{2\sum_{t=1}^{T} \frac{\lambda_{t;T}^4}{(\lambda_{t;T}^2+\alpha_T)^2}}} \xrightarrow{d} N(0;1) \tag{12}$$

provided $\alpha_T \sum_{t=1}^{T} \frac{\lambda_{t;T}^4}{(\lambda_{t;T}^2+\alpha_T)^2} \to \infty$ and the assumptions for asymptotic normality of the efficient estimator hold.

Such a test is valuable when working with ICA systems as it provides a feedback about the validity of the entire structure.

4 Simulations

I generate samples of η_t through equation $\eta_t = \Theta\epsilon_t$ with various distributions for the epsilons and a sample size of $T = 150$. I compare the performance in recovering the lag polynomial of the efficient estimator of Section 4 to that of efficient GMM based on moment conditions (*i.e.* deriving identifying equations implied by independence under the assumptions that moments up to order 4 exists, see *e.g.*, Guay (2021)) and fastICA (Oja and Yuan, 2006). In the forthcoming tables, the corresponding estimators are denoted by log-cf, GMM, and fICA, respectively.

I consider the following distributions for the sources: student with 3 degrees of freedom, uniform on [-1;1], Binomial(20, 0.3), and Gamma(5, 1/7). All distributions are centered as to have mean zero. These distributions account for a variety of cases such as fat tails, skewness, or presence of atoms.

Consider first a student distribution with 3 degrees of freedom. In this case, the student distribution exhibits fat tails and moments higher than 2 do not exist, endangering identification strategies based on higher moments.

It appears the estimator based on log-empirical characteristic function considerably outperforms both efficient GMM and fastICA estimators when the sources are student distributed. The gains come mostly from a lower standard deviation, though there is some bias reduction especially compared to fastICA.

Now, I turn to uniform and binomial distributions. Both distribution have all their moments but one is continuous and symmetric while the other is discrete and

skewed. Both efficient GMM and the characteristic-function based estimator outperform fastICA for the uniform distribution. In the binomial case, the characteristic function based estimator again fares better than both efficient GMM and fastICA, with a considerable reduction in mean square error originating from lower standard deviations.

5 Application to SVAR

5.1 Structural Vector Autoregression

Structural Vector Autoregressions (SVAR) have attracted a lot of interest in time series econometrics since the pioneering work of Sims (1980). The standard model postulates that some observed state of the economy characterized by a vector of n variables, Y_t, is related to unobserved (stationary) shocks (*e.g.*, monetary or oil shocks) through

$$Y_t = \Theta(L)\varepsilon_t \tag{13}$$

where $\Theta(L)$ is an unknown lag polynomial that represents the impulse response function[4]. $\Theta(L)$ describes the transmission mechanism of shocks to the economy and a subset of its column typically constitutes parameters of interest.

The first step towards estimation of $\Theta(L)$ is usually to perform the vector autoregression $A(L)Y_t = \eta_t$ to recover estimates of the innovation vector, η_t. The fundamentalness assumption states that the span of the shocks and innovations are identical and thus $\eta_t = \Theta\varepsilon_t$ for some invertible matrix Θ. It is well-known in the literature (see for instance Forni, Gambetti and Sala (2019)) that Θ corresponds to the first term in the lag polynomial $\Theta(L)$.

Once Θ is identified, the whole lag polynomial is recovered as $A(L)^{-1}\Theta$. As a result, the problem is reduced to the system $\eta_t = \Theta\varepsilon_t$. While standard SVAR only assumes that entries in ε_t are uncorrelated, second moments – $\Sigma_\eta = \Theta\Sigma_\varepsilon\Theta'$ – bring too few equations to solve for Θ. Various solutions have been proposed, among which

[4]Similarly to ICA systems, shocks as well as their effects on the system are unobserved, which leads to a scale indeterminacy: shocks can be arbitrarily re-scaled to get an observationally equivalent system in which their effects are inversely re-scaled. The unit variance normalization (each shock has variance one) or the unit effect normalization ($\Theta_{jj} = 1\ \forall j$) are frequently imposed.

short-run restrictions (Sims, 1980), long-run restrictions (King et al., 1987; Blanchard and Quah, 1989; Shapiro and Watson, 1988), identification by heteroskedasticity (Rigobon, 2003; Sentana and Fiorentini, 2001; Lewis, 2019), or sign restrictions (Uhlig, 2005). A good recent reference is Kilian and Lütkepohl (2017).

Although these restrictions solve the identification problem, assuming a priori knowledge of numerous $(n(n-1))$ shocks' effects is often an issue, as the transmission mechanism of shocks to the economy is primarily an empirical question. Thus, some authors (Siegfried et al., 2002; Gourieroux, Monfort et al., 2014) have pointed out that $\eta_t = \Theta\varepsilon_t$ can be identified by assuming that ε_t contains non-Gaussian independent variables, bypassing restrictions on Θ and introducing ICA methods to the SVAR literature. Many subsequent studies have followed that road, estimating the model using high-order moments (Guay, 2021; Keweloh, 2019), or pseudo-maximum of likelihood (Gouriéroux, Monfort and Renne, 2017). See also related discussions and methods in Moneta et al. (2013); Herwartz (2015); Lanne, Meitz and Saikkonen (2017).

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6 Appendix A: proofs

6.1 Theorem 3.1 (Consistency)

Proof. Consistency follows by verifying assumptions of Theorem 2.1 in Newey and McFadden (1994). The parameter space is compact by construction, identification is established, and the limiting objective function is continuous by dominated convergence since characteristic functions are bounded. It remains to show uniform convergence in probability.

The empirical characteristic function using a consistent estimator of η_t converges uniformly in probability:

$$
\begin{aligned}
\sup_{\Theta\in\overline{\Theta}}\left|\frac{1}{T}\sum_{t=1}^{T}e^{i\vec{s}P_j\hat{\eta}_t}-\mathbb{E}[e^{i\vec{s}P_j\eta_t}]\right| &= \sup_{\Theta\in\overline{\Theta}}\left|\frac{1}{T}\sum_{t=1}^{T}e^{i\vec{s}P_j\hat{\eta}_t}-\frac{1}{T}\sum_{t=1}^{T}e^{i\vec{s}P_j\eta_t}+\frac{1}{T}\sum_{t=1}^{T}e^{i\vec{s}P_j\eta_t}-\mathbb{E}[e^{i\vec{s}P_j\eta_t}]\right| \\
&\leq \sup_{\Theta\in\overline{\Theta}}\left|\frac{1}{T}\sum_{t=1}^{T}e^{i\vec{s}P_j\hat{\eta}_t}-\frac{1}{T}\sum_{t=1}^{T}e^{i\vec{s}P_j\eta_t}\right|+\sup_{\Theta\in\overline{\Theta}}\left|\frac{1}{T}\sum_{t=1}^{T}e^{i\vec{s}P_j\eta_t}-\mathbb{E}[e^{i\vec{s}P_j\eta_t}]\right| \\
&\leq \sup_{\Theta\in\overline{\Theta}}\left|\frac{1}{T}\sum_{t=1}^{T}e^{i\vec{s}P_j\eta_t}(e^{i\vec{s}P_j(\hat{\eta}_t-\eta_t)}-1)\right|+o_p(1) \\
&\leq \sup_{\Theta\in\overline{\Theta}}\frac{1}{T}\sum_{t=1}^{T}\left|e^{i\vec{s}P_j(\hat{\eta}_t-\eta_t)}-1\right|+o_p(1) \\
&\leq \sup_{\Theta\in\overline{\Theta}}\frac{1}{T}\sum_{t=1}^{T}|\vec{s}P_j(\hat{\eta}_t-\eta_t)|+o_p(1) \\
&\leq \frac{1}{T}\sum_{t=1}^{T}\|\vec{s}\|\frac{1}{\cos(\frac{\pi}{2}-\epsilon)}|\hat{\eta}_t-\eta_t|+o_p(1) \\
&\to^p 0
\end{aligned}
$$

where the second inequality follows from the uniform law of large numbers, the fourth from $|e^{ix}-1|\leq|x|$, the fifth from ϵ-invertibility, and the convergence is implied by consistency of $\hat{\eta}_t$.

Hence, by Theorem 2.1 in Newey and McFadden (1994), $\hat{\theta}\to^p\theta_0$. □

6.2 Theorem 3.2 (Asymptotic Normality)

Proof. Derivation of asymptotic normality follows the approach in Newey and McFadden (1994). The arguments of integration $\vec{s}$, $\vec{r}$ do not play a significant role and are suppressed to ease notation. By dominated convergence, first-order conditions read

$$\int \left(\Re \frac{\partial q_T}{\partial \theta}(\hat{\theta}) \quad \Im(\frac{\partial q_T}{\partial \theta}(\hat{\theta})) \right) W_T \begin{pmatrix} \Re q_T(\hat{\theta}) \\ \Im q_T(\hat{\theta}) \end{pmatrix} = 0 \tag{14}$$

Applying the mean-value theorem around the true value θ_0 yields

$$\int \left(\Re \frac{\partial q_T}{\partial \theta}(\hat{\theta}) \quad \Im(\frac{\partial q_T}{\partial \theta}(\hat{\theta})) \right) W_T \begin{pmatrix} \Re q_T(\theta_0) + (\Re \frac{\partial q_T}{\partial \theta}(\tilde{\theta}_R))'(\hat{\theta} - \theta_0) \\ \Im q_T(\theta_0) + (\Im \frac{\partial q_T}{\partial \theta}(\tilde{\theta}_I))'(\hat{\theta} - \theta_0) \end{pmatrix} = 0$$

so that, rearranging

$$\sqrt{T}(\hat{\theta} - \theta_0) = - \left[\int \left(\Re \frac{\partial q_T}{\partial \theta}(\hat{\theta}) \quad \Im(\frac{\partial q_T}{\partial \theta}(\hat{\theta})) \right) W_T \begin{pmatrix} (\Re \frac{\partial q_T}{\partial \theta}(\tilde{\theta}_R))' \\ (\Im \frac{\partial q_T}{\partial \theta}(\tilde{\theta}_I))' \end{pmatrix} \right]^{-1}$$
$$\sqrt{T} \int \left(\Re \frac{\partial q_T}{\partial \theta}(\hat{\theta}) \quad \Im(\frac{\partial q_T}{\partial \theta}(\hat{\theta})) \right) W_T \begin{pmatrix} \Re q_T(\theta_0) \\ \Im q_T(\theta_0) \end{pmatrix}$$

The first term can be proven to converge uniformly in probability, proceeding as in the uniform convergence step in proving consistency. For the second term, the empirical characteristic function converges to a complex normal stochastic process. This follows from convergence of finite dimensional distributions (by the multivariate central limit theorem and the delta method) and tightness (see Feuerverger, Mureika et al. (1977)[56]).

Then convergence of $\sqrt{T}(\frac{1}{T}\sum_{i=1}^{T} e^{i\vec{v}\eta_t} - \mathbb{E}[e^{i\vec{v}\eta_t}])$ to a complex normal stochastic process together with the continuous mapping theorem deliver asymptotic normality.

[5] As pointed out by Csorgo (1981), the result requires slightly stronger conditions than initially thought. Existence of moments larger than 1 suffices.

[6] They also proved almost sure convergence of $\sup_{-K \leq s \leq K} |c_n(s) - c(s)|$, where c denotes the empirical characteristic function and c_n its empirical counterpart, the empirical characteristic function is almost surely bounded away from zero on a neighborhood of the origin for T large enough, and tightness can thus be established for the cumulant generating function as well. As a result, the log-version of the criterion can also be shown to be asymptotically normal

If η_t is known, we directly obtain

$$\sqrt{T}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0; BVB')$$

Interestingly, estimation of η_t does not affect the asymptotic variance. Indeed, estimation of η_t can be accounted for by expanding $q_0(\hat{\eta}_t) = q_0(\eta_t(\hat{\beta}))$ into $q_0(\eta_t(\beta_0)) + \frac{\partial q(\theta_0, \eta_t(\beta_0))}{\partial \beta}(\hat{\beta} - \beta)$ plus lower-order terms, where β is the underlying parameter vector in estimating η_t. The first term corresponds to the case where η_t is observed. If η_t is an error term independent from its regressors x_t, then by properties of the P_j,

$$\begin{aligned}
\frac{\partial q_0}{\partial \beta} &= \varphi(\vec{s}) \sum_{j=0}^{n} a_j \frac{\mathbb{E}[e^{i\vec{s}P_j\eta_t}(-x_t)P_j'\vec{s}']}{\mathbb{E}[e^{i\vec{s}P_j\eta_t}]} \\
&= -\varphi(\vec{s})\mathbb{E}[x_t] \left(\sum_{j=0}^{n} a_j P_j' \right) \vec{s}' \\
&= 0
\end{aligned}$$

□

Appendix B: Additional results

The asymptotic variance

Due to the asymptotic linear representation shown while establishing asymptotic normality, the variance can be approximated via bootstrap. Alternatively, the asymptotic variance can be consistently estimated. Indeed, most terms appearing in its expression have a natural estimator based on the use of $\hat{\theta}$ in place of θ_0 and the use of the sample counterparts of population quantities.

Assume for ease of exposition that the characteristic functions do not vanish. This simplifies slightly the following formulae by allowing characteristic functions in the denominators, but is not necessary. We have

$$\frac{\partial q_0}{\partial \theta'} = -\varphi(\vec{s}) \sum_{j=1}^{n} \frac{\partial \ln(\mathbb{E}[e^{i\vec{s}P_j\eta_t}])}{\partial \theta'} = -\varphi(\vec{s}) \sum_{j=1}^{n} \frac{\mathbb{E}\left[e^{i\vec{s}P_j\eta_t}(\eta_t' \otimes \vec{s})\frac{\partial \operatorname{vec}(P_j)}{\partial \theta'}\right]}{\mathbb{E}[e^{i\vec{s}P_j\eta_t}]} \tag{15}$$

As an illustration of the differentiated term, consider the two-dimensional case:

$$\Theta = \begin{pmatrix} \cos(\theta_1) & \cos(\theta_2) \\ \sin(\theta_1) & \sin(\theta_2) \end{pmatrix} \tag{16}$$

Tedious but straightforward algebra yields

$$\Theta^{-1} = \frac{1}{\sin(\theta_2 - \theta_1)} \begin{pmatrix} \sin(\theta_2) & -\cos(\theta_2) \\ -\sin(\theta_1) & \cos(\theta_1) \end{pmatrix} \tag{17}$$

$$P_1 = \frac{1}{\sin(\theta_2 - \theta_1)} \begin{pmatrix} \cos(\theta_1)\sin(\theta_2) & -\cos(\theta_1)\cos(\theta_2) \\ \sin(\theta_1)\sin(\theta_2) & -\sin(\theta_1)\cos(\theta_2) \end{pmatrix} \tag{18}$$

$$P_2 = \frac{1}{\sin(\theta_2 - \theta_1)} \begin{pmatrix} -\sin(\theta_1)\cos(\theta_2) & \cos(\theta_1)\cos(\theta_2) \\ -\sin(\theta_1)\sin(\theta_2) & \cos(\theta_1)\sin(\theta_2) \end{pmatrix} \tag{19}$$

$$\frac{\partial \operatorname{vec}(P_1)}{\partial \theta'} = \frac{1}{\sin^2(\theta_2 - \theta_1)} \begin{pmatrix} \sin(\theta_2)\cos(\theta_2) & -\sin(\theta_1)\cos(\theta_1) \\ \sin^2(\theta_2) & -\sin^2(\theta_1) \\ -\cos^2(\theta_2) & \cos^2(\theta_1) \\ -\sin(\theta_2)\cos(\theta_2) & \sin(\theta_1)\cos(\theta_1) \end{pmatrix} \tag{20}$$

$$\frac{\partial \operatorname{vec}(P_2)}{\partial \theta'} = \frac{\partial \operatorname{vec}(I - P_1)}{\partial \theta'} = -\frac{\partial \operatorname{vec}(P_1)}{\partial \theta'} \tag{21}$$

Hence, replacing expectations by sample averages and using consistent estimators in place of unknown parameters delivers a consistent estimate of $\frac{\partial q_0}{\partial \theta'}$.

Regarding the central term, the (centered) empirical characteristic function converges to a mean zero process with covariance function $\varphi(\vec{s}+\vec{r}) - \varphi(\vec{s})\varphi(\vec{r})$. The covariance functions is enough to characterize the complex process noting that $\overline{\varphi(\vec{s})} = \varphi(-\vec{s})$.

Letting $\Phi(\vec{s}) \stackrel{\text{def}}{=} \begin{pmatrix} \varphi(\vec{s}) \\ \varphi(\vec{s}_1) \\ ... \\ \varphi(\vec{s}_n) \end{pmatrix}$, where $\vec{s}_j \stackrel{\text{def}}{=} \vec{s}P_j$, and $A \stackrel{\text{def}}{=} \begin{pmatrix} a_0 & a_1 & ... & a_n \end{pmatrix}$, the covariances are obtained from

$$\begin{aligned} &\mathbb{C}ov\left(\begin{pmatrix} \Re q_0(\vec{u}) \\ \Im q(\theta_0, \vec{u}) \end{pmatrix}; \begin{pmatrix} \Re q_0(\vec{v}) \\ \Im q(\theta_0, , \vec{v}) \end{pmatrix}\right) \\ &= \begin{pmatrix} A & 0 \\ 0 & A \end{pmatrix} \mathbb{C}ov\left(\begin{pmatrix} \Re\Phi(\vec{u}) \\ \Im\Phi(\vec{u}) \end{pmatrix}; \begin{pmatrix} \Re\Phi(\vec{v}) \\ \Im\Phi(\vec{v}) \end{pmatrix}\right) \begin{pmatrix} A' & 0 \\ 0 & A' \end{pmatrix} \end{aligned}$$

where, using properties of the complex-normal distribution,

$$\mathbb{C}ov(\Re\varphi(\vec{u}_j); \Re\varphi(\vec{v}_k)) = \frac{1}{2}\Re\left(\frac{\varphi(\vec{u}_j - \vec{v}_k)}{\varphi(\vec{u}_j)\varphi(-\vec{v}_k)} + \frac{\varphi(\vec{u}_j + \vec{v}_k)}{\varphi(\vec{u}_j)\varphi(\vec{v}_k)} - 2\right) \tag{22}$$

$$\mathbb{C}ov(\Re\varphi(\vec{u}_j); \Im\varphi(\vec{v}_k)) = \frac{1}{2}\Im\left(-\frac{\varphi(\vec{u}_j - \vec{v}_k)}{\varphi(\vec{u}_j)\varphi(-\vec{v}_k)} + \frac{\varphi(\vec{u}_j + \vec{v}_k)}{\varphi(\vec{u}_j)\varphi(\vec{v}_k)}\right) \tag{23}$$

$$\mathbb{C}ov(\Im\varphi(\vec{u}_j); \Re\varphi(\vec{v}_k)) = \frac{1}{2}\Im\left(\frac{\varphi(\vec{u}_j - \vec{v}_k)}{\varphi(\vec{u}_j)\varphi(-\vec{v}_k)} + \frac{\varphi(\vec{u}_j + \vec{v}_k)}{\varphi(\vec{u}_j)\varphi(\vec{v}_k)} - 2\right) \tag{24}$$

$$\mathbb{C}ov(\Im\varphi(\vec{u}_j); \Im\varphi(\vec{v}_k)) = \frac{1}{2}\Re\left(\frac{\varphi(\vec{u}_j - \vec{v}_k)}{\varphi(\vec{u}_j)\varphi(-\vec{v}_k)} - \frac{\varphi(\vec{u}_j + \vec{v}_k)}{\varphi(\vec{u}_j)\varphi(\vec{v}_k)}\right) \tag{25}$$

All terms can be consistently estimated by using empirical characteristic functions as estimates of their population counterparts.

Remark: The covariance kernel is easier to handle directly in complex form. Let $\phi_j(\vec{s}) \stackrel{\text{def}}{=} \prod_{m\neq j} \hat{\varphi}(\vec{s}P_m)$ with $\phi_0(\vec{s}) = 1$. The covariance kernel is given by $K(\vec{r}, \vec{s}) = \sum_{k=0}^{n}\sum_{j=0}^{n} a_k a_j \phi_j(\vec{s})\phi_k(-\vec{r})(\varphi(\vec{r}_k - \vec{s}_j) - \varphi(\vec{r}_k)\varphi(\vec{s}_j))$. This is also valid

when a characteristic function vanishes.

Remark: Similar results can be derived for the log version of the condition, *i.e.*, $\sum_{j=0}^{n} a_j \ln\left(\frac{1}{T}\sum_{t=1}^{T} e^{i\vec{s}P_j\eta_t}\right) = 0$. The centered empirical log-characteristic function converges to a zero mean process with covariance function $\frac{\varphi(\vec{s}+\vec{r})}{\varphi(\vec{s})\varphi(\vec{r})} - 1$ and satisfies $\overline{\ln(\varphi(\vec{s}))} = \ln(\varphi(-\vec{s}))$. The covariance kernel is then given by $\sum_{k=0}^{n}\sum_{j=0}^{n} a_k a_j \frac{\varphi(\vec{r}_k-\vec{s}_j)}{\varphi(\vec{r}_k)\varphi(-\vec{s}_j))}$.

Eigenfunctions

I derive the eigenfunctions and eigenvalues in details. Consider the estimated counterpart of the integral operator K:

$$
\begin{aligned}
(\hat{K}g)(\vec{r}) &= \int \begin{pmatrix} \widehat{\mathbb{C}ov}(\Re q_T(\vec{s}), \Re q_T(\vec{r}))g_1(\vec{s}) + \widehat{Cov}(\Im q_T(\vec{s}), \Re q_T(\vec{r}))g_2(\vec{s}) \\ \widehat{\mathbb{C}ov}(\Re q_T(\vec{s}), \Im q_T(\vec{r}))g_1(\vec{s}) + \widehat{\mathbb{C}ov}(\Im q_T(\vec{s}), \Im q_T(\vec{r}))g_2(\vec{s}) \end{pmatrix} \, \mathrm{d}\vec{s} \\
&= \sum_{k=0}^{n}\sum_{j=0}^{n} \frac{a_k a_j}{2} \\
&\times \int \begin{pmatrix} \Re\left(\frac{\hat{\varphi}(\vec{s}_j-\vec{r}_k)}{\hat{\varphi}(\vec{s}_j)\hat{\varphi}(-\vec{r}_k)} + \frac{\hat{\varphi}(\vec{s}_j+\vec{r}_k)}{\hat{\varphi}(\vec{s}_j)\hat{\varphi}(\vec{r}_k)} - 2\right) g_1(\vec{s}) + \Im\left(\frac{\hat{\varphi}(\vec{s}_j-\vec{r}_k)}{\hat{\varphi}(\vec{s}_j)\hat{\varphi}(-\vec{r}_k)} + \frac{\hat{\varphi}(\vec{s}_j+\vec{r}_k)}{\hat{\varphi}(\vec{s}_j)\hat{\varphi}(\vec{r}_k)}\right) g_2(\vec{s}) \\ \Im\left(-\frac{\hat{\varphi}(\vec{s}_j-\vec{r}_k)}{\hat{\varphi}(\vec{s}_j)\hat{\varphi}(-\vec{r}_k)} + \frac{\hat{\varphi}(\vec{s}_j+\vec{r}_k)}{\hat{\varphi}(\vec{s}_j)\hat{\varphi}(\vec{r}_k)}\right) g_1(\vec{s}) + \Re\left(\frac{\hat{\varphi}(\vec{s}_j-\vec{r}_k)}{\hat{\varphi}(\vec{s}_j)\hat{\varphi}(-\vec{r}_k)} - \frac{\hat{\varphi}(\vec{s}_j+\vec{r}_k)}{\hat{\varphi}(\vec{s}_j)\hat{\varphi}(\vec{r}_k)}\right) g_2(\vec{s}) \end{pmatrix} \, \mathrm{d}\vec{s} \\
&= \frac{1}{T}\sum_{\tau=1}^{T}\sum_{k=0}^{n}\sum_{j=0}^{n} \frac{a_k a_j}{2} \\
&\times \int \begin{pmatrix} \Re\left(\frac{e^{i(\vec{s}_j-\vec{r}_k)\eta_\tau}}{\hat{\varphi}(\vec{s}_j)\hat{\varphi}(-\vec{r}_k)} + \frac{e^{i(\vec{s}_j+\vec{r}_k)\eta_\tau}}{\hat{\varphi}(\vec{s}_j)\hat{\varphi}(\vec{r}_k)} - 2\right) g_1(\vec{s}) + \Im\left(\frac{e^{i(\vec{s}_j-\vec{r}_k)\eta_\tau}}{\hat{\varphi}(\vec{s}_j)\hat{\varphi}(-\vec{r}_k)} + \frac{e^{i(\vec{s}_j+\vec{r}_k)\eta_\tau}}{\hat{\varphi}(\vec{s}_j)\hat{\varphi}(\vec{r}_k)}\right) g_2(\vec{s}) \\ \Im\left(-\frac{e^{i(\vec{s}_j-\vec{r}_k)\eta_\tau}}{\hat{\varphi}(\vec{s}_j)\hat{\varphi}(-\vec{r}_k)} + \frac{e^{i(\vec{s}_j+\vec{r}_k)\eta_\tau}}{\hat{\varphi}(\vec{s}_j)\hat{\varphi}(\vec{r}_k)}\right) g_1(\vec{s}) + \Re\left(\frac{e^{i(\vec{s}_j-\vec{r}_k)\eta_\tau}}{\hat{\varphi}(\vec{s}_j)\hat{\varphi}(-\vec{r}_k)} - \frac{e^{i(\vec{s}_j+\vec{r}_k)\eta_\tau}}{\hat{\varphi}(\vec{s}_j)\hat{\varphi}(\vec{r}_k)}\right) g_2(\vec{s}) \end{pmatrix} \, \mathrm{d}\vec{s} \\
&= \frac{1}{T}\sum_{\tau=1}^{T}\sum_{k=0}^{n}\sum_{j=0}^{n} a_k a_j \\
&\times \int \begin{pmatrix} \Re\left(\left(\frac{e^{-i\vec{r}_k\eta_\tau}}{\hat{\varphi}(-\vec{r}_k)} + \frac{e^{i\vec{r}_k\eta_\tau}}{\hat{\varphi}(\vec{r}_k)}\right)\frac{e^{i\vec{s}_j\eta_\tau}}{2\hat{\varphi}(\vec{s}_j)} - 1\right) g_1(\vec{s}) + \Im\left(\frac{e^{-i\vec{r}_k\eta_\tau}}{\hat{\varphi}(-\vec{r}_k)} + \frac{e^{i\vec{r}_k\eta_\tau}}{\hat{\varphi}(\vec{r}_k)}\right)\frac{e^{i\vec{s}_j\eta_\tau}}{2\hat{\varphi}(\vec{s}_j)} g_2(\vec{s}) \\ \Im\left(-\frac{e^{-i\vec{r}_k\eta_\tau}}{\hat{\varphi}(-\vec{r}_k)} + \frac{e^{i\vec{r}_k\eta_\tau}}{\hat{\varphi}(\vec{r}_k)}\right)\frac{e^{i\vec{s}_j\eta_\tau}}{2\hat{\varphi}(\vec{s}_j)} g_1(\vec{s}) + \Re\left(\left(\frac{e^{-i\vec{r}_k\eta_\tau}}{\hat{\varphi}(-\vec{r}_k)} - \frac{e^{i\vec{r}_k\eta_\tau}}{\hat{\varphi}(\vec{r}_k)}\right)\frac{e^{i\vec{s}_j\eta_\tau}}{2\hat{\varphi}(\vec{s}_j)}\right) g_2(\vec{s}) \end{pmatrix} \, \mathrm{d}\vec{s} \\
&= \frac{1}{T}\sum_{\tau=1}^{T}\sum_{k=0}^{n}\sum_{j=0}^{n} a_k a_j \int \begin{pmatrix} \left(\Re\frac{e^{i\vec{r}_k\eta_\tau}}{\hat{\varphi}(\vec{r}_k)}\Re\frac{e^{i\vec{s}_j\eta_\tau}}{\hat{\varphi}(\vec{s}_j)} - 1\right) g_1(\vec{s}) + \Re\frac{e^{i\vec{r}_k\eta_\tau}}{\hat{\varphi}(\vec{r}_k)}\Im\frac{e^{i\vec{s}_j\eta_\tau}}{\hat{\varphi}(\vec{s}_j)} g_2(\vec{s}) \\ \Im\frac{e^{i\vec{r}_k\eta_\tau}}{\hat{\varphi}(\vec{r}_k)}\Im i\frac{e^{i\vec{s}_j\eta_\tau}}{\hat{\varphi}(\vec{s}_j)} g_1(\vec{s}) + \Im\frac{e^{i\vec{r}_k\eta_\tau}}{\hat{\varphi}(\vec{r}_k)}\Re(-i)\frac{e^{i\vec{s}_j\eta_\tau}}{\hat{\varphi}(\vec{s}_j)} g_2(\vec{s}) \end{pmatrix} \, \mathrm{d}\vec{s} \\
&= \frac{1}{T}\sum_{\tau=1}^{T}\sum_{k=0}^{n}\sum_{j=0}^{n} a_k a_j \\
&\times \int \begin{pmatrix} \left(\Re\frac{e^{i\vec{r}_k\eta_\tau}}{\hat{\varphi}(\vec{r}_k)} - 1\right)\left(\Re\frac{e^{i\vec{s}_j\eta_\tau}}{\hat{\varphi}(\vec{s}_j)} - 1\right) g_1(\vec{s}) + \left(\Re\frac{e^{i\vec{r}_k\eta_\tau}}{\hat{\varphi}(\vec{r}_k)} - 1\right)\left(\Im\frac{e^{i\vec{s}_j\eta_\tau}}{\hat{\varphi}(\vec{s}_j)} - 1\right) g_2(\vec{s}) \\ \left(\Im\frac{e^{i\vec{r}_k\eta_\tau}}{\hat{\varphi}(\vec{r}_k)} - 1\right)\left(\Re\frac{e^{i\vec{s}_j\eta_\tau}}{\hat{\varphi}(\vec{s}_j)} - 1\right) g_1(\vec{s}) + \left(\Im\frac{e^{i\vec{r}_k\eta_\tau}}{\hat{\varphi}(\vec{r}_k)} - 1\right)\left(\Im\frac{e^{i\vec{s}_j\eta_\tau}}{\hat{\varphi}(\vec{s}_j)} - 1\right) g_2(\vec{s}) \end{pmatrix} \, \mathrm{d}\vec{s} \\
&= \frac{1}{T}\sum_{t=1}^{T} \begin{pmatrix} \Re\sum_{k=0}^{n} a_k \left(\frac{e^{it_k\eta_\tau}}{\hat{\varphi}(t_k)} - 1\right) \\ \Im\sum_{k=0}^{n} a_k \left(\frac{e^{it_k\eta_\tau}}{\hat{\varphi}(t_k)} - 1\right) \end{pmatrix} \\
&\times \left[\int \Re\sum_{j=0}^{n} a_j \left(\frac{e^{i\vec{s}_j\eta_\tau}}{\hat{\varphi}(\vec{s}_j)} - 1\right) g_1(\vec{s}) \, \mathrm{d}\vec{s} + \Im\sum_{j=0}^{n} a_j \left(\frac{e^{i\vec{s}_j\eta_\tau}}{\hat{\varphi}(\vec{s}_j)} - 1\right) g_2(\vec{s}) \, \mathrm{d}\vec{s}\right]
\end{aligned}
$$

noting that $\frac{1}{T}\sum_{\tau} \frac{e^{i\vec{r}_k\eta_\tau}}{\hat{\varphi}(\vec{r}_k)} = 1$.

This implies that the eigenfunctions g_1 and g_2 take the form

$$g_1(\vec{r}) = \frac{1}{T}\sum_{\tau=1}^{T} c_\tau \Re \sum_{k=0}^{n} a_k \left(\frac{e^{i\vec{r}_k \eta_\tau}}{\hat{\varphi}(\vec{r}_k)} - 1 \right) \tag{26}$$

and

$$g_2(\vec{r}) = \frac{1}{T}\sum_{\tau=1}^{T} c_\tau \Im \sum_{k=0}^{n} a_k \left(\frac{e^{i\vec{r}_k \eta_\tau}}{\hat{\varphi}(\vec{r}_k)} - 1 \right) \tag{27}$$

Substituting these in the system $(\hat{K}g)(\vec{r}) = \mu g(\vec{r})$, it follows that the coefficients $\{c_\tau^m, \tau = 1, \ldots, T\}$ for $m = 1, \ldots, T$, form the T eigenvectors of the matrix M with elements

$$M_{\tilde{\tau}\tau} = \Re \int \sum_{j=0}^{n} a_j \left(\frac{e^{i\vec{s}_j \eta_\tau}}{\frac{1}{T}\sum_{t=1}^{T} e^{i\vec{s}P_j \eta_t}} - 1 \right) \overline{\sum_{k=0}^{n} a_k \left(\frac{e^{i\vec{s}_k \eta_{\tilde{\tau}}}}{\frac{1}{T}\sum_{t=1}^{T} e^{i\vec{s}P_k \eta_t}} - 1 \right)} \, \mathrm{d}\vec{s}$$

The associated eigenvalues correspond to $T\mu_{m;T}$.

Note:

- M/T has eigenvalues μ, eigenvectors E.

$\hat{\phi} = \frac{1}{T}\sum_{\tau=1}^{T} c_\tau (\sum_{k=0}^{n} a_k (\frac{e^{i\vec{s}P_k \eta_\tau}}{T^{-1}\sum_{\tau}^{T} e^{i\vec{s}P_k \eta_\tau}} - 1))/\|\phi\| = E'B/T/\|\phi\|$ is a a normalized eigenfunction, using E as the eigenvector. B is the base, i.e., the collection of $(\sum_{k=0}^{n} a_k (\frac{e^{i\vec{s}P_k \eta_\tau}}{T^{-1}\sum_{\tau}^{T} e^{i\vec{s}P_k \eta_\tau}} - 1))$.

- $\|\phi\| = \sqrt{E'ME/T} = \sqrt{\mu}$

- $< \hat{\phi}, f > = \int f E'B/T/\sqrt{\mu}\Pi(\mathrm{d}\vec{s})$

- $Q = 1/T^2 \sum_\tau \frac{1}{\mu_\tau^2 + \alpha} |E'_\tau \int f\overline{B}|^2$.

Now, I derive the eigenfunctions and eigenvalues in details. Consider the integral operator $\hat{K}$ applied to $g : \mathbb{R}^n \to \mathbb{R}^2$

$$(Kg)(\vec{r}) = \int \begin{pmatrix} \mathbb{C}ov(\Re q_T(\vec{s}), \Re q_T(\vec{r}))g_1(\vec{s}) + \mathbb{C}ov(\Im q_T(\vec{s}), \Re q_T(\vec{r}))g_2(\vec{s}) \\ \mathbb{C}ov(\Re q_T(\vec{s}), \Im q_T(\vec{r}))g_1(\vec{s}) + \mathbb{C}ov(\Im q_T(\vec{s}), \Im q_T(\vec{r}))g_2(\vec{s}) \end{pmatrix} \, \mathrm{d}\vec{s}$$

Doing the algebra on its estimated counterpart yields

$$\begin{aligned}(\hat{K}g)(\vec{r}) = \frac{1}{T}\sum_{t=1}^{T} &\begin{pmatrix} \Re \sum_{k=0}^{n} a_k \left(\frac{e^{it_k\eta_\tau}}{\hat{\varphi}(t_k)} - 1 \right) \\ \Im \sum_{k=0}^{n} a_k \left(\frac{e^{it_k\eta_\tau}}{\hat{\varphi}(t_k)} - 1 \right) \end{pmatrix} \\ &\times \left[\int \Re \sum_{j=0}^{n} a_j \left(\frac{e^{i\vec{s}_j\eta_\tau}}{\hat{\varphi}(\vec{s}_j)} - 1 \right) g_1(\vec{s}) \, \mathrm{d}\vec{s} + \Im \sum_{j=0}^{n} a_j \left(\frac{e^{i\vec{s}_j\eta_\tau}}{\hat{\varphi}(\vec{s}_j)} - 1 \right) g_2(\vec{s}) \, \mathrm{d}\vec{s} \right]\end{aligned}$$

This implies that the eigenfunctions g_1 and g_2 take the form

$$g_1(\vec{r}) = \frac{1}{T}\sum_{\tau=1}^{T} c_\tau \Re \sum_{k=0}^{n} a_k \left(\frac{e^{i\vec{r}_k\eta_\tau}}{\hat{\varphi}(\vec{r}_k)} - 1 \right) \tag{28}$$

and

$$g_2(\vec{r}) = \frac{1}{T}\sum_{\tau=1}^{T} c_\tau \Im \sum_{k=0}^{n} a_k \left(\frac{e^{i\vec{r}_k\eta_\tau}}{\hat{\varphi}(\vec{r}_k)} - 1 \right) \tag{29}$$

where the coefficients, $\{c_\tau^m, \tau = 1, \ldots, T\}$ for $m = 1, \ldots, T$, form the T eigenvectors of the matrix M with elements

$$M_{\tilde{\tau}\tau} = \Re \int \sum_{j=0}^{n} a_j \left(\frac{e^{i\vec{s}_j\eta_\tau}}{\frac{1}{T}\sum_{t=1}^{T} e^{i\vec{s}P_j\eta_t}} - 1 \right) \overline{\sum_{k=0}^{n} a_k \left(\frac{e^{i\vec{s}_k\eta_{\tilde{\tau}}}}{\frac{1}{T}\sum_{t=1}^{T} e^{i\vec{s}P_k\eta_t}} - 1 \right)} \, \mathrm{d}\vec{s} \tag{30}$$

The associated eigenvalues correspond to $T\mu_{m;T}$.

The scalar products in computing the objective function then read

$$
\begin{aligned}
< q_T, \varphi_{m;T} > &= \int \left(\Re \sum_{j=0}^{n} a_j \ln \left(\frac{1}{T} \sum_{t=1}^{T} e^{i\vec{s}P_j\eta_t} \right) \quad \Im \sum_{j=0}^{n} a_j \ln \left(\frac{1}{T} \sum_{t=1}^{T} e^{i\vec{s}P_j\eta_t} \right) \right) \\
&\quad \begin{pmatrix} \frac{1}{T} \sum_{\tau=1}^{T} c_\tau^m \Re \sum_{k=0}^{n} a_k \left(\frac{e^{i\vec{r}_k\eta_\tau}}{\frac{1}{T}\sum_{t=1}^{T} e^{i\vec{s}P_k\eta_t}} - 1 \right) \\ \frac{1}{T} \sum_{\tau=1}^{T} c_\tau^m \Im \sum_{k=0}^{n} a_k \left(\frac{e^{i\vec{r}_k\eta_\tau}}{\frac{1}{T}\sum_{t=1}^{T} e^{i\vec{s}P_k\eta_t}} - 1 \right) \end{pmatrix} \, \mathrm{d}\vec{s} \\
&= \frac{1}{T} \sum_{\tau=1}^{T} c_\tau^m \, \Re \int \sum_{j=0}^{n} a_j \ln \left(\frac{1}{T} \sum_{t=1}^{T} e^{i\vec{s}P_j\eta_t} \right) \overline{\sum_{k=0}^{n} a_k \left(\frac{e^{i\vec{r}_k\eta_\tau}}{\frac{1}{T}\sum_{t=1}^{T} e^{i\vec{s}P_k\eta_t}} - 1 \right)} \, \mathrm{d}\vec{s}
\end{aligned}
$$

Further useful expressions:

$$\left\langle \frac{\partial q_T}{\partial \theta}, \sum_{k=0}^{n} a_k \phi_k(\vec{s}) e^{i\vec{s}P_k\eta_\tau} \right\rangle$$
$$= i \sum_{k=0}^{n}\sum_{j=1}^{n} \frac{\partial vec(P_j)'}{\partial \theta} \frac{1}{T} \sum_{t=1}^{T} \left(\eta_t \otimes \left(\frac{1}{T^n} \sum e^{-\frac{1}{2}\|\eta_{\tilde{t}} + \sum_{m=1}^{n} P_m \eta_{t_m}\|^2} \left(\eta_{\tilde{t}} + \sum_{m=1}^{n} P_m \eta_{t_m} \right) \right) \right)$$
$$- i \sum_{k=0}^{n}\sum_{j=1}^{n} \frac{\partial vec(P_j)'}{\partial \theta} \frac{1}{T} \sum_{t=1}^{T} \left(\eta_t \otimes \left(\frac{1}{T} \sum_{\tilde{t}=1}^{T} e^{-\frac{1}{2}\|\eta_{\tilde{t}} + P_j\eta_t + P_k\eta_\tau\|^2} \left(\eta_{\tilde{t}} + P_j\eta_t + P_k\eta_\tau \right) \right) \right)$$

noting that $\int \vec{s} e^{\vec{s}(a+bi)} \pi(\mathrm{d}\vec{s}) = (a+bi)\ e^{\|a\|^2/2} e^{a'bi} e^{-\|b\|^2/2}$.

$$\left\langle \frac{\partial q_T}{\partial \theta'}, q_T^{\tau} \right\rangle$$
$$= \left\langle \frac{\partial \frac{1}{T}\sum_{t=1}^{T} \left(e^{i\vec{s}\eta_t} - \prod_{j=1}^{n} \hat{\varphi}_\eta(\vec{s}P_j) \right)}{\partial \theta'}, \left(e^{i\vec{s}\eta_\tau} - \prod_{j=1}^{n} \hat{\varphi}_\eta(\vec{s}P_j) \right) \right\rangle$$
$$= -\int \sum_{j=1}^{n} \prod_{\{k\neq j\}} \hat{\varphi}_\eta(\vec{s}P_k) \hat{\mathbb{E}}[\exp(i\vec{s}P_j\eta_{t_j}) i(\eta'_{t_j} \otimes \vec{s})] \frac{\partial \mathrm{vec}(P_j)}{\partial \theta'} \overline{\left(e^{i\vec{s}\eta_\tau} - \prod_{j=1}^{n} \hat{\varphi}_\eta(\vec{s}P_j) \right)} \pi(\mathrm{d}\vec{s})$$
$$= \sum_{j=1}^{n} \frac{-1}{T^n} \sum_{t_1,\cdots,t_n} \left[\int e^{i\vec{s}(P_j\eta_{t_j} + \sum_{k\neq j} P_k\eta_{t_k} - \eta_\tau)} (\eta'_{t_j} \otimes \vec{s}) i \right.$$
$$\left. - \frac{1}{T^n} \sum_{\tau_1,\cdots,\tau_n} e^{i\vec{s}(P_j\eta_{t_j} + \sum_{k\neq j} P_k\eta_{t_k} - \sum_{m=1}^{n} P_m\eta_{\tau_m})} (\eta'_{t_j} \otimes \vec{s}) i\pi(\vec{s}) \right] \frac{\partial \mathrm{vec}(P_j)}{\partial \theta'}$$
$$= \sum_{j=1}^{n} \frac{1}{T^{2n}} \sum_{t_1,\cdots,t_n,\tau_1,\cdots,\tau_n} \left(\eta'_{t_j} \otimes \right.$$
$$\left. \left[\left(\sum_{k=1}^{n} P_k\eta_{t_k} - \eta_\tau \right)' e^{-0.5\|\sum_{k=1}^{n} P_k\eta_{t_k} - \eta_\tau\|^2} - \left(\sum_{k=1}^{n} P_k\eta_{t_k} - \sum_{m=1}^{n} P_m\eta_{\tau_m} \right)' e^{-0.5\|\sum_{k=1}^{n} P_k\eta_{t_k} - \sum_{m=1}^{n} P_m\eta_{\tau_m}\|^2} \right] \right)$$
$$\frac{\partial \mathrm{vec}(P_j)}{\partial \theta'}$$

where $\hat{\varphi}_\eta(\vec{s}) = \frac{1}{T}\sum_{t=1}^{T} e^{i\vec{s}\eta_t}$

$$
\begin{aligned}
&\partial_\theta \langle q_T, q_T^\tau \rangle \\
&= \sum_{m=1}^{n} \left(\frac{\partial \mathrm{vec}(P_m)}{\partial \theta'} \right)' \\
&\left\{ \sum_{t_1,\ldots,t_n} \frac{(\eta_{t_m} \otimes I)}{T^n} \left[\left(\sum_{k=1}^{n} P_k \eta_{t_k} - \eta_\tau \right) e^{-\|\sum_{k=1}^{n} P_k \eta_{t_k} - \eta_\tau\|^2/2} + \frac{1}{T} \sum_{\tau=1}^{T} \left(\sum_{k=1}^{n} P_k \eta_{t_k} - \eta_\tau \right) e^{-\|\sum_{k=1}^{n} P_k \eta_{t_k} - \eta_\tau\|^2/2} \right] \right. \\
&\left. - \frac{1}{T^{2n}} \sum_{t_1,\ldots,t_n} \sum_{\tau_1,\ldots,\tau_n} ((\eta_{t_k} - \eta_{\tau_k}) \otimes I) \left(\sum_{k=1}^{n} P_k \eta_{t_k} - \sum_{m=1}^{n} P_m \eta_{\tau_m} \right) e^{-\|\sum_{k=1}^{n} P_k \eta_{t_k} - \sum_{m=1}^{n} P_m \eta_{\tau_m}\|^2/2} \right\}
\end{aligned}
$$

$$
\begin{aligned}
&\partial_\theta \langle q_T, q_T^\tau \rangle \\
&= \sum_{m=1}^{n} \left(\frac{\partial \mathrm{vec}(P_m)}{\partial \theta'} \right)' \\
&\left\{ \frac{1}{T^n} \sum_{t_1,\ldots,t_n} \left[(\eta_{t_m} \otimes I) \left(\sum_{k=1}^{n} P_k \eta_{t_k} - \eta_\tau \right) e^{-0.5\|\sum_{k=1}^{n} P_k \eta_{t_k} - \eta_\tau\|^2} \right. \right. \\
&\qquad \left. \left. + \frac{1}{T} \sum_{\tau=1}^{T} (\eta_{t_m} \otimes I) \left(\sum_{k=1}^{n} P_k \eta_{t_k} - \eta_\tau \right) e^{-0.5\|\sum_{k=1}^{n} P_k \eta_{t_k} - \eta_\tau\|^2} \right] \right. \\
&\left. - \frac{1}{T^{2n}} \sum_{t_1,\ldots,t_n} \sum_{\tau_1,\ldots,\tau_n} ((\eta_{t_k} - \eta_{\tau_k}) \otimes I) \left(\sum_{k=1}^{n} P_k \eta_{t_k} - \sum_{m=1}^{n} P_m \eta_{\tau_m} \right) e^{-0.5\|\sum_{k=1}^{n} P_k \eta_{t_k} - \sum_{m=1}^{n} P_m \eta_{\tau_m}\|^2} \right\}
\end{aligned}
$$

$$
\begin{aligned}
&\langle \partial_\theta q_T, q_T^\tau \rangle \\
&= \sum_{m=1}^{n} \left(\frac{\partial \mathrm{vec}(P_m)}{\partial \theta'} \right)' \\
&\left\{ \frac{1}{T} \sum_{\tau=1}^{T} (\eta_{t_m} \otimes I) \left(\sum_{k=1}^{n} P_k \eta_{t_k} - \eta_\tau \right) e^{-0.5\|\sum_{k=1}^{n} P_k \eta_{t_k} - \eta_\tau\|^2} \right. \\
&\left. - \frac{1}{T^{2n}} \sum_{t_1,\ldots,t_n} \sum_{\tau_1,\ldots,\tau_n} (\eta_{t_k} \otimes I) \left(\sum_{k=1}^{n} P_k \eta_{t_k} - \sum_{m=1}^{n} P_m \eta_{\tau_m} \right) e^{-0.5\|\sum_{k=1}^{n} P_k \eta_{t_k} - \sum_{m=1}^{n} P_m \eta_{\tau_m}\|^2} \right\}
\end{aligned}
$$

7 Polar coordinates

For simplicity, assume that θ is organized in matrix form: θ is an $(n-1) \times n$ matrix whose k-th column characterizes that of Θ. Then,

$$\Theta_{\cdot k} = \begin{pmatrix} \cos(\theta_{nk}) \cos(\theta_{n-1,k}) \dots \cos(\theta_{2k}) \cos(\theta_{1k}) \\ \cos(\theta_{nk}) \cos(\theta_{n-1,k}) \dots \cos(\theta_{2k}) \sin(\theta_{1k}) \\ \cos(\theta_{nk}) \cos(\theta_{n-1,k}) \dots \sin(\theta_{2k}) \\ \dots \\ \cos(\theta_{nk}) \cos(\theta_{n-1,k}) \\ \cos(\theta_{nk}) \sin(\theta_{n-1,k}) \\ \sin(\theta_{nk}) \end{pmatrix}$$